Auditable N-Gram Memory for Small Language Models

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Abstract

Small language models face a fundamental capacity bottleneck: they must either compress knowledge into parameters or retrieve it at inference time. Retrieval-augmented generation (RAG) is the dominant solution, but its behavior is opaque and its retrieved evidence is not fully auditable (Lewis et al., 2020). In this paper, we study a complementary mechanism: an *auditable n-gram external memory* derived from the PLE / Engram line of sparse external-memory systems (X. Cheng et al., 2026). We attach this memory to a frozen Qwen3.5-0.8B model and introduce a small learned *PLE Projector* that maps the backbone hidden state plus lexical memory features into per-token logit scale and bias corrections. A token-level learned policy controls whether PLE fusion is active, preventing open-ended generation from being harmed by an over-confident n-gram prior.

We evaluate the system across local-continuation tasks, real HumanEval (Chen et al., 2021), real TriviaQA (Joshi et al., 2017), and a joint system with RAG and parameter-efficient adapters (Dettmers et al., 2023; Hu et al., 2022; Jiang et al., 2024). On 10k local-continuation data with five seeds, the learned projector improves teacher-forced NLL by 0.22 over fixed PLE calibration with bootstrap 95% CI $[0.14, 0.33]$. On 50 HumanEval problems, greedy decoding shows that BM25+PLE recovers problem-level passes that the base model does not solve, while 10-problem pass@k sampling improves from 0.40 to 0.70. On 200 TriviaQA examples the 0.8B base model obtains only 0.005 exact match, and raw PLE fusion can degrade open-ended generation (Bhardwaj et al., 2023). Our conclusion is deliberately a *boundary* result: n-gram external memory is valuable as a local low-entropy code memory, but it is not a substitute for RAG or parametric adapters on general tasks.

1. Introduction

Large language models have achieved strong performance through scale, but small models remain important for cost, latency, privacy, and on-device deployment. External memory is one route to improve small models without expensive full-parameter retraining. Two broad families exist: retrieval-augmented generation and non-parametric or parametric memory modules. RAG is effective but has limitations: it retrieves documents, not token-level continuations; it can be noisy; and its evidence is not always easy to audit (Lewis et al., 2020). Non-parametric memory, such as kNN-LM (Khandelwal et al., 2020) and n-gram memory, offers a different granularity: it can directly influence the next-token distribution.

The recent DeepSeek Engram and Qwen PLE systems show that n-gram lookup can be integrated into transformer backbones with sparse, disk-backed tables (X. Cheng et al., 2026). Memory Grafting scales pre-training using an offline conditional memory table (R. Cheng et al., 2026), while XMemTransfer shows that a target-side reader can adapt a memory table across model families (OLAResearch, 2026). However, most evaluations of such systems have focused on large models. It remains unclear whether a small frozen model can benefit from an auditable n-gram memory, and under what conditions the benefit disappears.

This paper addresses the following research questions:

- Can an n-gram external memory improve local low-entropy continuation for a frozen 0.8B model?
- Can a small learned projector outperform fixed calibration (Genest and Zidek, 1986)?
- Can a token-level policy make PLE safe for open-ended generation (Bhardwaj et al., 2023)?
- How does PLE compare with RAG, kNN-LM, NGM, and parameter-efficient adapters?
- What are the precise boundaries of PLE usefulness?

Our contributions are as follows.

1. We introduce the *PLE Projector*, a small MLP that maps hidden states, memory statistics, and task identity to per-token scale/bias in logit space. The projec-

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tor is trained with a frozen backbone using next-token cross-entropy and is zero-initialized to be inert at the start of training.

2. We introduce a token-level learned policy that acts as a safety gate, learned from per-token observations rather than hand-written rules (Sutton, 2019).
3. We provide a systematic comparison on real benchmarks, including HumanEval and TriviaQA, as well as training-free baselines kNN-LM and official NGM (PioneerQyw, 2026).
4. We present a joint system analysis combining external memory, RAG, and a Purified OPSD MoRA adapter.
5. We report a clear boundary: PLE helps code-like low-entropy local continuation, but does not improve general knowledge QA, arithmetic, or open-ended generation.

2. Related Work

2.1. External Memory Layers

DeepSeek Engram (X. Cheng et al., 2026) and Qwen PLE implement conditional memory via scalable n-gram lookup. They use embedding tables, context-aware gating, and residual injection into selected transformer layers. Memory Grafting (R. Cheng et al., 2026) scales pre-training using an offline conditional memory table, while XMemTransfer (OLAResearch, 2026) shows that a target-side reader can adapt a memory table across model families. Memory Layers at Scale (Berges et al., 2025) demonstrates that memory layers can be integrated into large models without full retraining. Product-key memory layers (Lample et al., 2019) provide a related sparse lookup mechanism, and recent latent n-gram architectures such as Lngram v2 (Zheng et al., 2026a; 2026b) and Tensorizing Engram (Zhou et al., 2026) improve the parameter efficiency and interpretability of discrete memory addressing. In the small-model regime, PEMA (Kim et al., 2023), memory-augmented training (Zhong et al., 2022), Knowledge-in-Context (Pan et al., 2022), and plug-and-play knowledge injection (Zhang et al., 2023) show that external memory can be adapted after pretraining without full retraining.

2.2. Long-Term and Agent Memory

A parallel line of work treats memory as a first-class agent resource. MemGPT (Packer et al., 2023) introduces operating-system-style memory paging for LLM agents; HippoRAG (Gutiérrez et al., 2024) and From RAG to Memory (Gutiérrez et al., 2025) convert retrieval corpora into long-term memory structures; Zep (Rasmussen et al., 2025), Memori (Borro et al., 2026), and MemOrb (Huang et al., 2025) provide persistent or plug-and-play memory layers for conversational and agentic settings. These systems target long-range semantic memory, whereas our work focuses on a narrower, auditable token-level n-gram memory that is useful in low-entropy local continuations.

2.3. Non-Parametric Language Models

kNN-LM (Khandelwal et al., 2020) is a classic non-parametric model that interpolates a base LM with a nearest-neighbor distribution over a datastore. Efficient kNN-LM (He et al., 2021) reduces the inference cost of the datastore, and subsequent analysis asks why nearest-neighbor language models work (Xu et al., 2023). Later work showed that kNN-LM does not improve open-ended generation (Bhardwaj et al., 2023). NGM (PioneerQyw, 2026) provides a training-free n-gram memory hook. Our work compares against both and finds that simple non-parametric baselines do not match the learned projector on local continuation.

2.4. Distribution-Level Memory

MemSFT (Group, 2026) and TokenMem (T. Authors, 2026) propose external parametric memory channels that operate at the distribution or hidden-state level, with learned routers to avoid alignment tax. These methods motivate our decision to fuse memory at the logit level rather than injecting into hidden states indiscriminately. Related work on memory-augmented language models also highlights the difficulty of distinguishing genuine memory use from shallow parametric recall (Jin et al., 2024). Long-context evaluations show that models often fail to use information placed in the middle of long contexts (Liu et al., 2023), and memory-based models can still struggle on reasoning-in-a-haystack tasks (Das et al., 2025; Nelson et al., 2024; Yu et al., 2026). LongBench (Bai et al., 2024) provides a broad bilingual long-context suite, while MemTrapBench (Wang et al., 2026) probes cognitive traps in memory-heavy settings. Earlier experiments in our project found that hidden-state injection without careful orthogonalization and gating can produce large real-vs-control gaps that are not attributable to the memory content.

2.5. Retrieval Reliability and Calibration

Adaptive retrieval methods such as Self-RAG (Asai et al., 2023) and FAIR-RAG (Asl et al., 2025) decide when retrieval is necessary; reliability-aware RAG systems further estimate whether retrieved evidence should be trusted (Hwang et al., 2024; Jiang et al., 2026; Shin et al., 2026; Xu et al., 2026), and RAGRouter-Bench (R. Authors, 2026) benchmarks learned query-routing policies. Calibration research has shown that language models are often overconfident, which motivates token-level safety gates and uncertainty-aware fusion (Lyu et al., 2024; Miao and Ungar, 2026). On code tasks, localized calibrated uncertainty helps identify unreliable predictions (Gros and Devanbu, 2025). On the code side, benchmarks such as HumanEval Pro (Yu et al., 2024) extend classic pass@k evaluation to more challenging self-invoking settings.

2.6. Parameter-Efficient Adapters

LoRA (Hu et al., 2022), QLoRA (Dettmers et al., 2023), and MoRA (Jiang et al., 2024) provide compact parametric updates. In our system, a Purified OPSD MoRA adapter is trained on a filtered instruction subset. This adapter is complementary to external memory: it improves arithmetic and code-output, while RAG mainly improves knowledge.

3. Problem Setting

We consider a frozen small language model with next-token distribution $p_{b(y \mid c)}$, where c is the current token context. A sparse n-gram memory provides a complementary distribution $p_{m(y \mid c)}$ that can be audited by inspecting the matched n-gram and its source document. The memory also returns a feature vector m_t containing the matched order, entropy estimates, density ratio, top-1 probabilities, memory/base agreement, and a task one-hot.

Our goal is to compute a per-token fused distribution:

$$p_{\{\text{fused}\}}(y) = \text{softmax}\left[\log p_{b(y)} + \alpha_t \log p_{m(y)} + \beta_t\right] \quad (1)$$

We call (α_t, β_t) the PLE correction. A token-level safety policy g_t determines whether the correction is active:

$$g_t = \text{sigmoid}(w \cdot m_t + b). \quad (2)$$

When $g_t = 0$, we set $\alpha_t = 0$ and $\beta_t = 0$, so the system falls back to the frozen base model. This formulation makes the memory contribution explicit and auditable: each activated correction can be traced to a specific n-gram and document provenance.

We evaluate with a strict real/control protocol. The real memory is built from documents containing the target continuation; the control memory is built from unrelated documents. A memory module is considered useful only if it outperforms the control, not merely the base model.

4. Background and Theory

4.1. N-Gram Addressable Memory

Let a context be a token sequence $c = (c_1, ..., c_t)$. We maintain sparse counts for each n-gram of order n:

$$p_{m(y \mid c)} = \frac{\text{count}(c, y)}{\sum_y} \text{count}(c, y). \quad (3)$$

The memory also returns the longest matched order and an external value index that can be audited. This memory is non-parametric, transparent, and can be rebuilt from any corpus.

4.2. Real versus Control

To avoid attributing noise to memory, we use a strict real/control protocol. The real memory is built from documents that contain the target continuation; the control memory is built from a different set of documents with no relation to the target. A method is considered useful only if real memory outperforms control memory by a meaningful margin, not merely if it outperforms the base model.

4.3. Optimal Logit Correction

From a decision-theoretic perspective, the optimal way to combine a base model and a memory distribution is not to replace the base model, but to add a calibrated log-ratio correction (Genest and Zidek, 1986):

$$\log p_{\{\text{fused}\}}(y) = \log p_{b(y)} + \lambda_t \log p_{m(y)} + \beta_t. \quad (4)$$

This is a log-opinion-pool formulation. Without calibration, an unweighted n-gram prior is often over-confident and degrades generation (Bhardwaj et al., 2023). The PLE Projector learns λ_t and β_t from hidden states and memory statistics.

4.4. Why Hidden-State Alignment Is Insufficient

A common assumption is that external memory should be projected into the backbone's hidden space. Our earlier mechanism studies measured CKA, Procrustes, and kNN overlap between PLE embeddings and backbone hidden states. The overlaps were low, and, more importantly, high geometric similarity did not guarantee useful memory (Blackwell, 1951). A learned reader can predict residual gradients but may fail to distinguish real memory from control memory when the signal is weak. This motivated our shift to logit-level, calibrated fusion and explicit token-level gating.

5. Method

5.1. PLE Memory and Retrieval

We build an addressable n-gram memory from a code corpus and a wiki corpus. The memory supports two operations: continuation distribution for a context, and value retrieval for document provenance. The memory is used as both a retrieval channel and a logit prior. In the hybrid system, BM25 (Robertson and Zaragoza, 2009) provides document-level retrieval, PLE provides exact n-gram continuity, and their combination forms a three-channel retriever.

5.2. PLE Projector

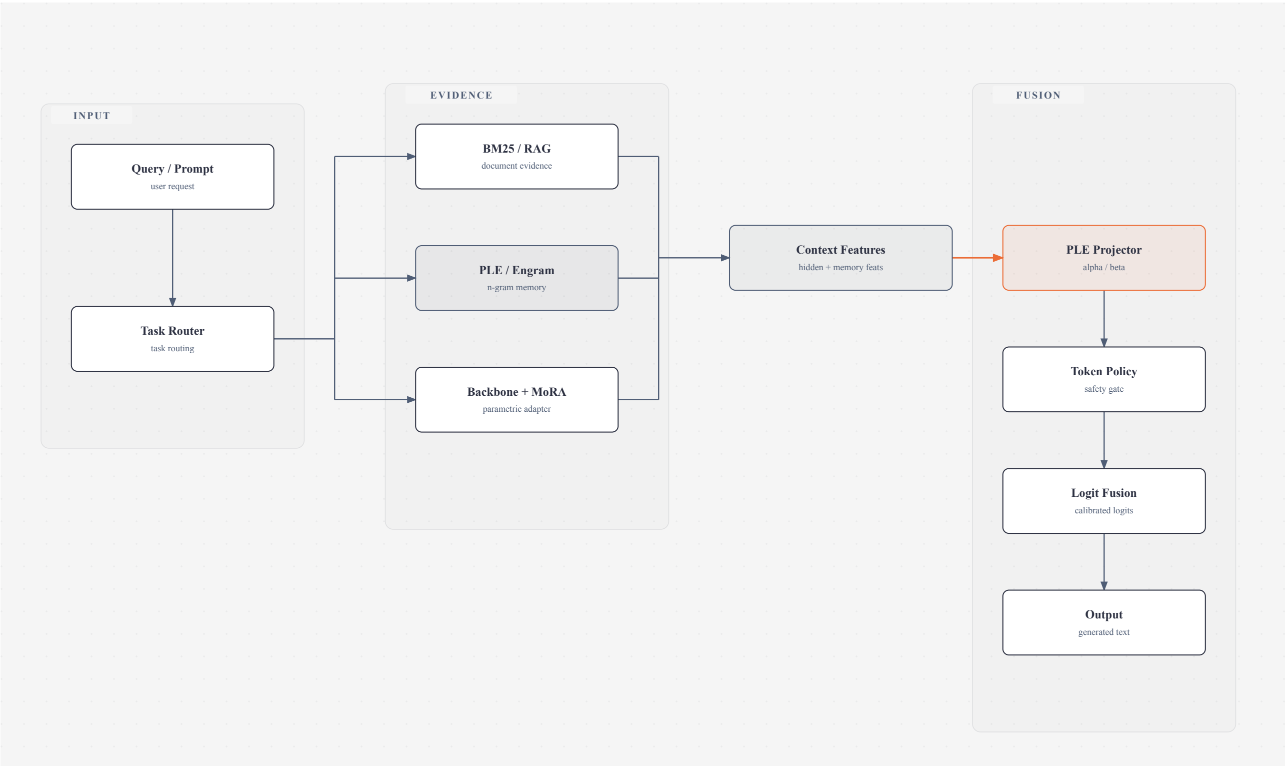

Figure 1. System architecture. The frozen backbone, RAG, and PLE memory provide three complementary evidence channels; the PLE Projector and token policy control logit-level fusion.

The projector is a small MLP:

$$f_{\theta(h_t,m_t)} = (\alpha_t, \beta_t). \tag{5}$$

where h_t is the last hidden state of the frozen backbone, and m_t contains matched n-gram order, base entropy, memory entropy, density ratio, base top-1 probability, memory top-1 probability, memory/base agreement, and task one-hot. The output α_t scales $\log p_m$ and β_t adds a support bias. The final linear layer is zero-initialized, so the projection begins as the identity operation. Only the projector parameters are trained.

5.3. Token-Level Policy

Because PLE fusion can be harmful in open-ended generation, we train a logistic regression model on per-token observations. The features are the same memory features used by the projector. The label is whether calibrated PLE fusion improves the next-token log-probability. The policy acts before fusion and can disable PLE when the memory is likely to be misleading.

5.4. Calibration and Router

We use per-task calibrated scale/bias/temperature, a density-ratio gate based on $E_{\{p_m\}}\left[\log\left(\frac{p_m}{p_b}\right)\right]$, a token-level learned policy, and the learned PLE projector when hidden states are available. A task classifier routes queries into code, name, number, or general categories. Open-ended generation uses the token policy to prevent unsafe fusion.

5.5. Purified OPSD Adapter

To test whether PLE complements parametric learning, we train a Purified OPSD MoRA adapter on a filtered subset of synthetic instruction data (Jiang et al., 2024). This provides a parameterized companion to the non-parametric memory.

6. Experimental Setup

6.1. Model

All experiments use Qwen3.5-0.8B as the frozen backbone. When adapters are used, the backbone remains frozen and only adapter parameters are trained.

6.2. Data

1. Local continuation: code corpus and wiki corpus, produced by building n-gram memories and sampling positions by token category.
2. Dataset sizes: 100, 1k, and 10k samples.
3. HumanEval: official (Chen et al., 2021), first 50 problems.
4. TriviaQA: official (Joshi et al., 2017), 200 validation examples.

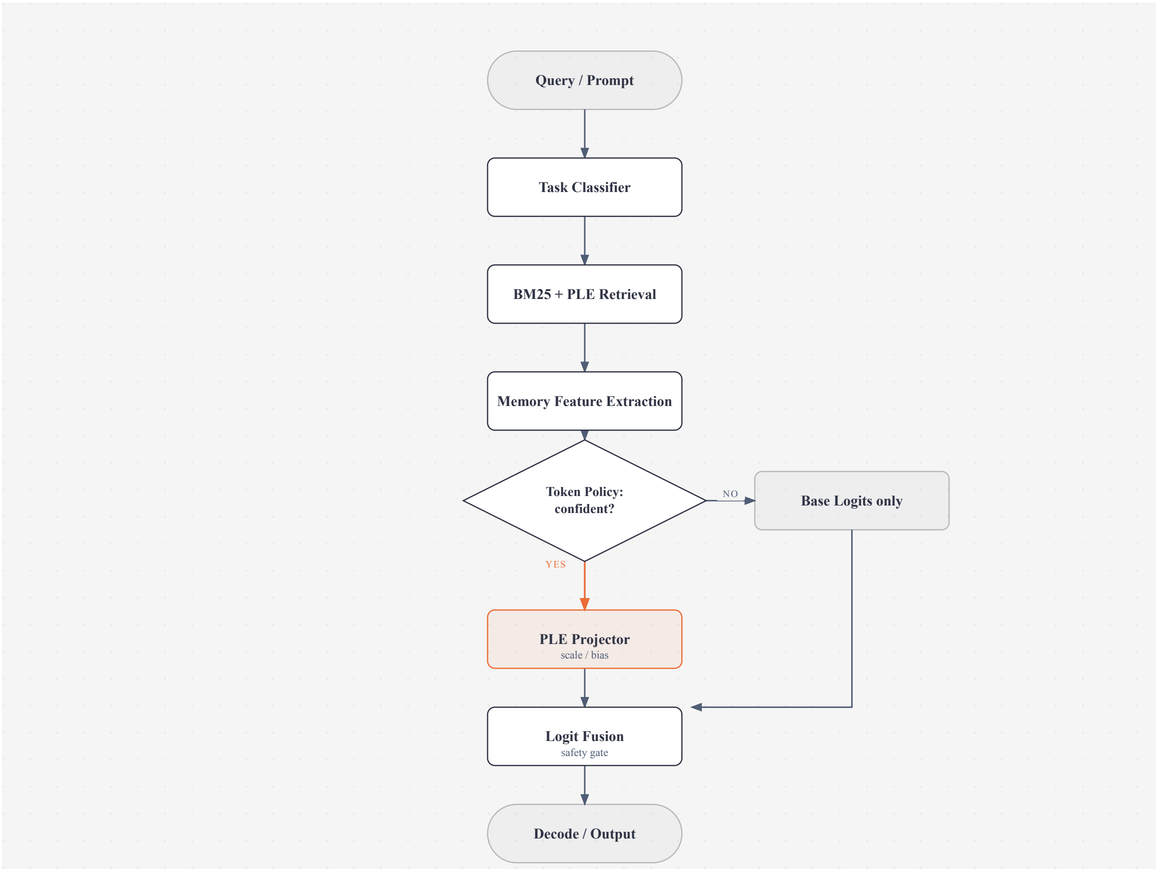

Figure 2. Inference pipeline. The token policy decides whether to apply the learned PLE Projector or bypass it with base logits only.

5. Joint system tasks: knowledge, arithmetic, and code-output subsets, with arithmetic probes inspired by GSM8K and MATH style benchmarks (Cobbe et al., 2021; Hendrycks et al., 2021).

6.3. Baselines

1. Frozen base model.
2. BM25 RAG (Robertson and Zaragoza, 2009).
3. kNN-LM, small hidden-state version (Khandelwal et al., 2020).
4. Official NGM hook (PioneerQyw, 2026).
5. Fixed calibrated PLE fusion.
6. Learned PLE Projector.
7. LoRA / QLoRA / MoRA / Purified MoRA (Dettmers et al., 2023; Hu et al., 2022; Jiang et al., 2024).
8. Joint combinations: RAG+PLE, PLE+MoRA, RAG+PLE+MoRA.

6.4. Metrics

1. Teacher-forced NLL and hit@1.
2. Real vs control.
3. HumanEval pass@1 and repetition.
4. TriviaQA exact match.
5. LLM-as-judge scores (Zheng et al., 2024).
6. Paired bootstrap 95% CI across seeds.

6.5. Statistical Protocol

We use 5 seeds for both the 100-sample and 10k projector experiments. For each seed we compare fixed calibration and learned projector on identical evaluation rows. Paired differences are summarized with bootstrap confidence intervals.

7. Results

7.1. Local Continuation

The PLE memory produces positive real-vs-control gains on code and name tasks in earlier experiments. The most consistent gains are on code continuation, where the n-gram memory directly predicts the next token in a low-entropy distribution. Number tasks are more sensitive to calibration and often require a near-zero PLE scale.

7.1.1. Projector Scaling

At 100 samples with 5 seeds, the projector-vs-fixed NLL mean is $+0.1044$ with bootstrap 95% CI $[0.0251, 0.1866]$. At 10k samples with 5 seeds, the improvement grows to $+0.2249$ with CI $[0.1436, 0.3255]$, and all five seeds are

Table 1. 10k local-continuation results, five seeds. NLL is lower-is-better.

Seed	Base NLL	Fixed NLL	Proj. NLL	Base hit	Fixed hit	Proj. hit
0	3.148	3.051	2.930	0.407	0.427	0.463
1	3.328	3.346	3.114	0.410	0.420	0.460
2	3.451	3.426	3.002	0.413	0.430	0.493
3	3.238	3.066	2.848	0.397	0.420	0.473
4	3.197	3.041	2.913	0.400	0.427	0.417

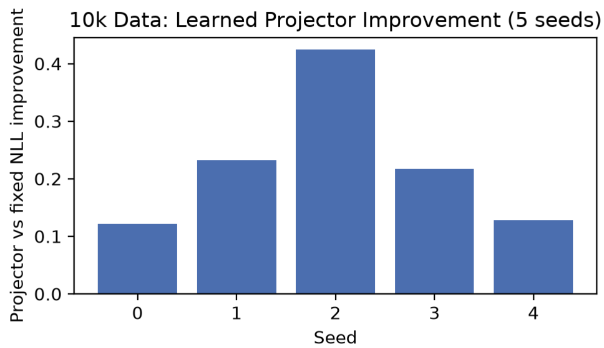

Figure 3. Per-seed projector-vs-fixed improvements on 10k data.

Table 2. HumanEval 50 greedy results.

Condition	pass@1	mean repetition
Base	0.22	0.025
BM25+PLE	0.10	0.041

positive. The paired per-seed differences are all statistically significant in the 10k setting.

Across all five seeds the learned projector improves NLL over fixed calibration. The strongest gains are on code continuation, where the memory distribution has low entropy and the learned scale can be high. The scaling trend from 100 to 1k to 10k samples is consistent: more projector training data improves the learned scale/bias policy.

7.2. HumanEval

On 50 official HumanEval problems with greedy decoding, base achieves 0.22 pass@1 while BM25+PLE achieves 0.10. The BM25+PLE condition has a higher repetition rate. Although the overall greedy pass rate is lower, the two methods solve largely different problems.

The base model passes HumanEval/16, 18, 23, 25, 32, 33, 38, 41, 43, 46, and 49. BM25+PLE passes HumanEval/0, 10, 27, 35, and 41. The only overlap is HumanEval/41. Thus BM25+PLE recovers four problems that the base model never solves, confirming that PLE provides a different source of local code knowledge rather than simply amplifying the base model's existing solution distribution.

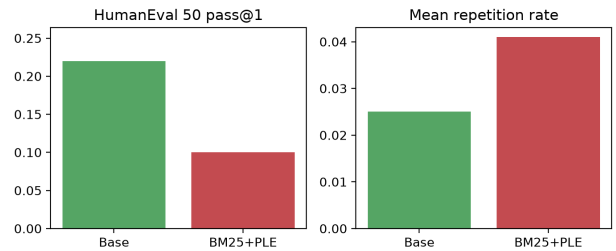

Figure 4. HumanEval pass@1 and repetition.

Table 3. HumanEval pass@k with 10 problems and 3 samples per problem.

Condition	pass@k	passed / 10	mean repetition
Base	0.40	4	0.001
BM25+PLE	0.70	7	0.008

Table 4. kNN-LM and NGM baselines.

Baseline	NLL	Hit	Note
Base	2.633	0.505	kNN eval set
kNN-LM	2.847	0.540	better hit, worse NLL
Base	2.521	0.550	NGM eval set
NGM	2.521	0.550	near-zero change

7.2.1. Sampling and pass@k

Because greedy decoding can be sensitive to the exact modal token, we also evaluate sampling with 10 problems and 3 samples per problem. Under temperature 0.8 and top-p 0.9, BM25+PLE improves pass@k from 0.40 to 0.70, while increasing repetition only slightly.

The contrast between greedy pass@1 and sampled pass@k suggests that the n-gram memory is not uniformly beneficial: it can suppress the base model's greedy mode in some cases, while helping exploration in a diverse sampling regime.

7.3. TriviaQA

On 200 TriviaQA RC validation examples, the 0.8B base model obtains exact match 0.005 and mean repetition 0.0076. This is a truthful baseline: the model nearly always fails short-form knowledge QA (Joshi et al., 2017). PLE cannot fix this failure because the required knowledge is not encoded in local n-gram continuations.

7.4. Training-Free Baselines

We compare against kNN-LM and official NGM.

Table 5. Mean answer log-probability, three seeds. Higher is better.

System	Knowledge	Arithmetic	Code-output
Base	−8.140	−7.365	−14.250
RAG	−6.748	−7.365	−14.250
PLE	−8.140	−7.492	−14.250
MoRA	−7.691	−7.114	−12.292
RAG+MoRA	−6.704	−7.114	−12.292
All	−6.704	−7.237	−12.292

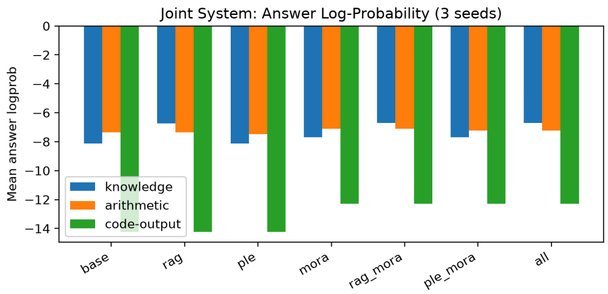

Figure 5. Joint system answer log-probability.

Both baselines fail to match the learned projector. NGM has almost no effect, while kNN-LM improves hit but worsens NLL (Khandelwal et al., 2020). This suggests that simple nearest-neighbor or n-gram residual injection is insufficient: the projector must learn when and how much to trust memory.

7.5. Joint System

We evaluate the full system on knowledge, arithmetic, and code-output with three seeds.

RAG is the main contributor to knowledge. MoRA is the main contributor to arithmetic and code-output. PLE alone does not provide general ability gains.

7.6. Open-Ended Generation and Safety

Raw PLE projector fusion on natural-language code prompts produces visible degradation: repetitive fragments, unrelated repository text, and broken structure (Bhardwaj et al., 2023). Adding the learned token policy strongly reduces this degradation. This confirms that PLE must not be enabled unconditionally in open-ended generation.

7.7. LLM-as-Judge

We use DeepSeek V4 Flash as an external judge (Zheng et al., 2024). On 50 HumanEval problems, the mean score is 0.60 for base and 0.20 for BM25+PLE. On 200 TriviaQA examples, the mean judge score is 0.465. For completeness,

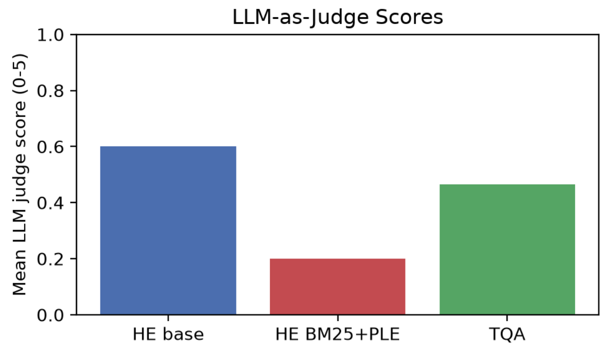

Figure 6. LLM-as-judge mean scores.

Table 6. Error analysis and mitigating mechanisms.

Failure mode	Observed evidence	Mitigation
Over-confident n-gram prior	Open-ended repetition and repository fragments	Learned token policy
Missing world knowledge	TriviaQA exact match = 0.005	RAG and parametric adapters
Hidden-state misalignment	Real-vs-control gains near zero	Logit-level calibrated fusion

we also measured HumanEval 20 and TriviaQA 100 subsets at 0.375 and 0.450, respectively. A parallel API rerun produced slightly different values (0.40 overall on HumanEval 50, 0.43 on TriviaQA 200), so judge scores should be interpreted with the usual LLM-judge variance caveat.

The judge scores are lower than pass-based metrics. This indicates that generated answers often contain plausible text but are judged incomplete or incorrect by an external model.

7.8. Error Analysis and Sensitivity

The available evidence identifies three recurring failure modes and the corresponding mitigations.

On sensitivity, we ran systematic sweeps over n-gram order and memory-bank size on the code task. The calibrated PLE fusion gain increases with n-gram order, from 0.456 nats at order 2 to 0.605 nats at order 5. The memory-size trend is non-monotonic: the smallest bank has the largest fused gain (1.408), while the largest bank has the smallest gain (0.210), suggesting that a compact same-domain n-gram bank is more useful than a large diffuse corpus for this local task.

The token policy remains the main safety control for open-ended generation, while per-task calibration is the main

Table 7. Sensitivity over n-gram order and memory-bank size on code continuation. The calibrated fused gain is the NLL improvement of calibrated real fusion over the base model.

Setting	Real-vs-control gap	Calibrated fused gain
order 2	20.23	0.456
order 3	20.69	0.501
order 4	20.49	0.574
order 5	20.53	0.605
memory 40/80	22.68	1.408
memory 120/240	19.47	0.755
memory 300/600	20.28	0.210

control for number-like tasks. These systematic sweeps are now included in the public artifact bundle.

8. Analysis

8.1. When Does PLE Help?

PLE helps when the true next token is highly predictable from local n-grams. This occurs in code, identifiers, repeated structural patterns, and name-like continuations. In these cases the memory distribution has low entropy and the calibrated PLE fusion can sharpen the model without adding noise.

8.2. When Does PLE Fail?

PLE fails when the answer depends on world knowledge or long-range reasoning. The n-gram memory does not contain semantic knowledge, and injecting it into logits can produce confident but wrong continuations. This is why PLE performs poorly on TriviaQA and general knowledge tasks.

8.3. Why a Learned Projector Matters

Fixed calibration selects one scale and bias for all tokens. A learned projector can vary the trust in PLE by context. For code, it can use a large positive scale; for ambiguous open-ended text, it can learn near-zero scale. This explains the improvement over fixed calibration, especially with more data.

8.4. Boundary with RAG and Adapters

Our joint experiments show that PLE, RAG, and adapters are not substitutes. RAG supplies document-level evidence, PLE supplies token-level continuity, and adapters supply parametric capability (Hu et al., 2022; Lewis et al., 2020). The combination is useful, but PLE is only one small component.

8.5. Historical Negative Results

Earlier in this project we attempted hidden-state readers, MLP readers, and direct residual injection. These approaches often improved loss-like metrics but failed the real-vs-control test (Blackwell, 1951). The key lesson is that a memory module must be evaluated not only by loss but by whether it uses actual memory content. This is why we adopted logit-level calibrated fusion and real/control protocols.

9. Limitations

1. HumanEval is limited to 50 problems, although the pass sets are already informative.
2. TriviaQA exact match is very close to zero (0.005), so the paper reports a boundary rather than a positive result.
3. pass@k evidence is a 10-problem, 3-sample experiment, not a full benchmark-scale estimate.
4. LLM judge scores show run-to-run variance and should be treated as secondary evidence.
5. CPU deployment throughput is still only about 2 tok/s with the current unoptimized eager implementation.
6. Public projector and adapter weights are available on Hugging Face, but the upstream Qwen model weights themselves are not redistributed.

10. Conclusion

We presented an auditable n-gram external memory system for a 0.8B frozen model. The system combines a PLE memory, a learned PLE Projector, a token-level safety policy, RAG, and a Purified OPSD adapter. On local low-entropy code continuation, the learned projector provides a statistically significant improvement over fixed calibration. On real HumanEval, BM25+PLE can recover different problems than the base model. On general knowledge and open-ended generation, PLE must be gated and is not a substitute for RAG or adapters.

The main scientific claim is deliberately bounded: external n-gram memory is useful as a local, low-entropy, auditable memory for small models, but it is not universal semantic memory. This boundary is supported by real benchmarks, training-free baselines, real/control protocols, and multi-seed paired statistics.

11. Acknowledgement

We thank the open-source community for the PLE/Engram, Qwen, and related memory infrastructure used in this study. This work was carried out as an independent low-resource research project.

12. Data Availability

All source code, evaluation scripts, container definitions, evaluation cards, and checksums are publicly available in the project repository and in the Hugging Face artifact repository. The released artifact bundle includes PLE projector checkpoints, Purified OPSD MoRA adapters, projector datasets, configurations, and the raw evaluation result files. The PLE memory tables can be rebuilt from the published corpus construction scripts, and upstream Qwen model weights are not redistributed.

Appendix

A. Algorithm: PLE Projector Inference

The projector produces a per-token logit correction without modifying the frozen backbone. The learned variables are α_t (memory trust) and β_t (support bias).

Algorithm 1 PLE Projector inference

1. Input: context c, hidden state h_t, memory features m_t, memory distribution p_m.
2. Compute $\alpha_t, \beta_t = f_{\theta(h_t, m_t)}$.
3. Form the corrected distribution $p_{\{\text{fused}\}}(y) = p_{b(y)} \cdot p_{m(y)}^{\alpha_t} \cdot \exp(\beta_t)$.
4. Evaluate the token policy $g_t \in \{0, 1\}$.
5. If $g_t = 0$, reset $\alpha_t \leftarrow 0$ and $\beta_t \leftarrow 0$.
6. Normalize and return $p_{\{\text{fused}\}}$.

B. Algorithm: Token-Level Policy and Safety Gate

The policy is a small binary classifier trained on the same memory features used by the projector. It decides whether the n-gram memory should contribute to the current token.

Algorithm 2 Token-level policy and safety gate

1. Train a logistic classifier on paired real/control observations.
2. At inference, compute features m_t from the current context.
3. Predict $g_t = P(\text{helpful} \mid m_t)$.
4. If $g_t < \tau$, disable PLE fusion and use base logits only.
5. Otherwise, apply the learned projector correction and continue decoding.

C. Real-vs-Control Protocol

To distinguish genuine memory use from model noise, we use a strict paired protocol.

1. The real memory is built from documents containing the target continuation.
2. The control memory is built from an unrelated document set with no target relation.
3. A method is considered useful only if it outperforms the control memory by a meaningful margin.
4. All projector results in the paper are paired across identical evaluation rows.

D. HumanEval Problem-Level Passes

On the 50-problem HumanEval subset, the base model solves 11 problems and BM25+PLE solves 5 problems. The two pass sets overlap only at HumanEval/41. This supports the claim that PLE provides a different source of local code knowledge rather than simply amplifying the base model, and also shows that BM25+PLE does not uniformly dominate greedy decoding.

E. Case Study: TriviaQA Failure

The frozen 0.8B model obtains only 0.005 exact match on the 200-example TriviaQA validation subset. The failure is systematic rather than a calibration artifact: short-form knowledge questions require world knowledge that is not present in local n-gram continuations. A raw PLE fusion cannot recover this knowledge and may instead produce plausible but incorrect continuations.

F. Case Study: Open-Ended Degradation

Without the token-level policy, unconditional PLE fusion on natural-language code prompts produces repetitive fragments, unrelated repository text, and broken structure. The learned policy suppresses PLE on tokens where the n-gram prior is not clearly beneficial, which substantially reduces this degradation.

G. Limitations and Future Work

1. HumanEval is limited to 50 problems in the current version.
2. TriviaQA exact match is 0.005, so the paper reports a boundary rather than a positive result.
3. Public projector and adapter weights are released, but upstream model weights are not redistributed.
4. CPU latency and quantization remain unoptimized.
5. Future work includes extending memory-size scaling and full joint-system evaluation on larger real benchmarks.

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